THE THREE REGULAR-IMMERSION FLAVOURS COINCIDE FOR IMMERSIONS INTO LOCALLY NOETHERIAN SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10564-10572 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 10574-10581, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Definitional-transfer template (same family as collected differential cards); the lane accepts written reductions even when one bridge lemma carries the weight. Feeds this wave's L16158/L16159 and the conormal card L16129's dictionary.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's definition section for regular-immersion flavours; Properties of Spaces, section on properties of spaces (locally Noetherian dictionary); Divisors, lemma regular-immersion-noetherian (the schemes equivalence).

Source reference. Original lemma 06BN: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10530–10546. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 063L: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10581–10581. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03E5: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10581–10581. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $i : Z \to X$ be an immersion of algebraic spaces over S. Assume X is locally Noetherian. Then i is Koszul-regular $\Leftrightarrow$ i is H_1-regular $\Leftrightarrow$ i is quasi-regular.*

Proof. Via Definition 06BN (and the definition of a locally Noetherian algebraic space in Properties of Spaces, Section 03E5) this immediately translates to the case of schemes which is Divisors, Lemma 063L. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.